

Classification and Clinical Forms of Leprosy

Leprosy may resolve spontaneously (in many cases) or progress into one of the clinical types described in the **Ridley-Jopling classification**, depending on the development of cell-mediated immunity (CMI). Another challenging group is **pure neural leprosy**, characterized by nerve involvement without skin lesions. Its reported frequency ranges from 1 to 10%, varying by geographic region and level of clinical awareness.

For field use, the **World Health Organization (WHO)** simplifies classification based on the number of skin lesions: - **Paucibacillary (PB) leprosy**: Five or fewer skin lesions. - **Multibacillary (MB) leprosy**: More than five skin lesions.

While this system is practical, studies indicate that up to 30% of patients may be misclassified as PB using this method alone, leading to potential undertreatment.

Treatment

The earliest documented treatment with modest efficacy was **chaulmoogra oil**, referenced in the *Sushruta Samhita* (600 BC). Though effects were limited, some benefit was observed in paucibacillary cases.

The first effective antimicrobial therapy emerged in 1943 with **promin**, an intravenous sulfone. This was soon followed by the introduction of **dapsone (diaminodiphenylsulfone, DDS)**, an oral sulfone that became the standard of care. However, by the 1970s, **dapsone resistance** became widespread. With the availability of **rifampicin (RMP)**, combination therapy was advocated to prevent resistance.

In 1982, the **WHO Chemotherapy Study Group** formally recommended multidrug therapy (MDT), combining **rifampicin, dapsone, and clofazimine**, depending on classification.

Today, **WHO-recommended multidrug therapy (MDT)** remains the global standard and is widely implemented.

Multidrug Therapy (MDT) Regimens

Paucibacillary Leprosy (PB)

- **Rifampicin:** 600 mg, once monthly under supervision for 6 months.
- **Dapsone:** 100 mg daily, unsupervised.

For a 60 kg adult, treatment can be discontinued after **6 supervised monthly doses administered within 9 months.**

A recent proposal suggests adding **clofazimine (Lamprene)** to the 6-month PB regimen to reduce undertreatment and harmonize therapy across classifications. This approach aims to minimize reliance on clinical classification, though it remains under debate among clinicians.

Multibacillary Leprosy (MB)

- **Monthly (supervised):** 600 mg rifampicin + 300 mg clofazimine (Lamprene).
- **Daily (unsupervised):** 100 mg dapsone + 50 mg clofazimine.

The official WHO recommendation is **12 supervised monthly doses within 18 months.** However, many clinicians modify this: - **Low bacterial index (BI):** 12 monthly doses over 18 months. - **High bacterial index ($BI \geq 4$):** 24 monthly doses over 36 months.

Doses are standardized for a 60 kg patient.

Uniform MDT (U-MDT)

The WHO is currently evaluating **Uniform MDT (U-MDT)** — a **6-month regimen using rifampicin, dapsone, and clofazimine for all leprosy**

patients, regardless of classification. This approach eliminates the need for classification.

However, this proposal is **controversial** and faces strong opposition from many clinicians due to concerns about undertreating multibacillary cases.

Treatment Outcomes and Safety Considerations

- The established MDT regimens have proven highly effective, with **relapse rates below 4%**.
- However, **follow-up duration for MB leprosy is often insufficient**, raising concerns about late relapses.
- **Dapsone-related haemolysis** is a known risk, particularly in **Nordic Caucasians**, even in the absence of G6PD deficiency. Reducing the dose to **50 mg daily** maintains efficacy while significantly reducing anaemia, likely due to genetic factors.
- **Dapsone hypersensitivity syndrome** is more common in **Nepalese and Chinese patients**, requiring heightened clinical vigilance.